# MagicCode Super-Resolution Software Release Notes v1.1.0

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## 1. Summary

MagicCode Super-Resolution Software v1.1.0 expands backend coverage, scaling flexibility, public processing modes, game engine integration, and runtime reporting for diagnostics and performance optimization.

## 2. Key Changes

- Expanded scaling range to 1.0x through 8.0x, subject to backend and model support.
- Public algorithm modes are now HIGH_SPEED_MODE and SPEED_MODE.
- Runtime backend selection supports MAGIC_BACKEND_X86, MAGIC_BACKEND_NEON, MAGIC_BACKEND_METAL, MAGIC_BACKEND_OPENGL, MAGIC_BACKEND_OPENGLES, MAGIC_BACKEND_VULKAN, and MAGIC_BACKEND_DEFAULT.
- Input types include CPU buffer input and GPU texture input types: INPUT_BUFFER, INPUT_TEXTURE_RGB8Unorm, and INPUT_TEXTURE_R8Unorm.
- Unity integration includes Android and iOS wrapper

support, C# bindings, and smoke validation workflow.

- Unreal Engine integration includes Blueprint/C++ bridge APIs and Android/iOS-oriented plugin configuration.
- Runtime status query exposes output dimensions, backend, input type, thread count, GPU time, and error code.
- Technical reporting now includes additional runtime and performance fields for diagnostics, such as report reason and processing statistics.

## 3. Compatibility

- Operating systems: Windows 10 64-bit and above, Android 8.0 and above, iOS 13.0 and above, iPadOS 13.0 and above, and macOS 12.0 and above.
- CPU architectures: x86_64 and ARM64.
- Required SIMD instruction sets: AVX2 for x86_64 paths and NEON for ARM64 paths.
- Input resolution range: minimum 64 x 64, maximum 4032 x 4032.
- Maximum CPU thread count: 8. Multi-threading is used when input height is at least 256 rows.

## 4. Migration Notes

- Replace old public mode names such as veryfast and fast with HIGH_SPEED_MODE and SPEED_MODE.
- Verify model files match the selected backend, mode, and scaling factor.
- Update integration code to populate the v1.1.0 input_param_t fields, especially input_type, backend, and log_level.
- Review the updated Privacy Policy and Data Reporting

Notice before distributing applications that include the SDK or plugins.

## 5. Known Limitations

- Not every backend supports every input type, scaling factor, or model combination.
- CPU backends use buffer input; GPU backends use texture input.
- Performance depends on device hardware, graphics driver, backend, model, input resolution, scaling factor, and integration method.
- Unreal Engine iOS packaging may require a compatible Xcode and Apple SDK version for the selected UE release.

## 6. Validation Snapshot

- Unity Android and iOS smoke tests passed with v1.1.0, producing valid 128x128 output from 64x64 input.
- Unreal Engine Android smoke test passed with native path MC_Init -> MC_Control(QUERY_STATUS) -> MC_Process -> MC_Uninit.
- Unreal Engine iOS bridge configuration is included; local buildability depends on the installed UE/Xcode toolchain compatibility.